ARSPI-Net: Neuromorphic Reservoir Dynamics for Affective-EEG State-Space Encoding

Andrew A. Lane*, Wendy Tang*, and Brady D. Nelson†

*Department of Electrical and Computer Engineering, Stony Brook University, Stony Brook, NY, USA

†Department of Psychology, Stony Brook University, Stony Brook, NY, USA

***Abstract*—Affective EEG event-related potentials (ERPs) are noisy, subject-dependent measurements of neural response dynamics. In this setting, robustness rankings can depend as much on the evaluation protocol as on the representation being evaluated. We compare four affective-ERP representations under subject-grouped validation: spectral band-power, ERP-window amplitudes, a fixed leaky integrate-and-fire reservoir summarized by a six-bin spike-count (BSC_6) embedding, and a trained EEGNet. The analysis focuses on amplitude-noise and channel-dropout perturbations. Two channel-dropout artifacts are quantified. First, raw retention favors near-chance models: the reservoir has 99% raw retention at 30% dropout largely because its clean balanced accuracy is only 0.463, close to the $1/3$ chance floor. Above-chance retention gives a more comparable measure. Second, zero-imputation of dropped channels creates out-of-distribution standardized values for non-centered features. We show algebraically that zero-imputation after standardization is not invariant to an information-preserving re-centering. Across 10–50% dropout and mean, kNN, and spatial fills, in-distribution imputation raises non-centered encoders by up to 0.11 balanced accuracy and changes the fixed-encoder ranking: ERP-window features overtake the reservoir, while the reservoir's zero-centered embedding changes little. For EEGNet, dropout fragility is mostly removed by training-time channel-dropout augmentation, which increases dropout balanced accuracy from 0.554 to 0.674 while preserving clean performance (0.70–0.71). The imputation effect also appears on DEAP, where the measured robustness of the same band-power features changes by up to 0.11 balanced accuracy under the imputation rule alone. These results support a simple reporting standard for EEG robustness studies: report above-chance retention, specify and vary the imputation rule, and state whether trained models used perturbation augmentation.**

***Index Terms*—EEG, event-related potentials, spiking neural networks, reservoir computing, affective computing, perturbation analysis.**

I. Introduction

Electroencephalographic event-related potentials (ERPs) provide millisecond-scale measurements of neural response dynamics, but the recorded waveform is a noisy, low-dimensional, and subject-dependent projection of an unobserved neural process [12], [13]. Inter-subject variability, trial averaging, volume conduction, and measurement noise can make clean-split accuracy difficult to interpret. A representation may separate affective conditions under one protocol while preserving little usable structure after amplitude noise, temporal misalignment, or channel loss.

Endpoint accuracy alone also says little about what a representation preserves. This issue is especially relevant for spiking encodings. Fixed recurrent reservoirs and spiking networks provide nonlinear temporal expansions [1], [4], but EEG/ERP studies often treat them primarily as classifiers [9], [15]. In that setting, the reservoir's discrete spike-count observables are not usually examined as perturbation-sensitive temporal measurements tied to post-stimulus windows.

The perturbation behavior of affective-ERP reservoir encodings remains underreported, even though preprocessing, acquisition, and electrode-loss choices can materially change EEG representation performance [16]–[18].

This paper evaluates whether channel-dropout robustness rankings for affective-ERP representations are stable under reasonable changes to the scoring protocol. We compare spectral band-power, ERP-window amplitudes, a fixed leaky integrate-and-fire (LIF) reservoir summarized by six post-onset spike-count bins (BSC_6) and compressed for a linear readout, and a trained EEGNet (Fig. 1). The reservoir is not treated as a proposed winner; it is used as a fixed reference whose embedding is nearly zero-centered and whose dynamics are not retrained under perturbation. Graph topology, structure–function coupling, and closed-loop analyses are outside this submission. The contributions are:

1) A subject-grouped robustness characterization of affective-ERP representations under amplitude noise and channel dropout, with per-fold confidence intervals and a permutation null (Sec. V-A, Tables I–II).
2) Quantification of two protocol artifacts in channel-dropout evaluation: raw-retention inflation near the chance floor, and zero-imputation bias for non-centered features. The imputation effect is also tested on DEAP (Secs. III-E, V-B, V-C).
3) Evidence that corrected imputation changes the fixed-encoder ranking, while channel-dropout augmentation largely accounts for the trained EEGNet's recovered dropout performance (Fig. 4, Table II).
4) A fixed-reservoir reference analysis, including a single-bin ablation, to separate the reservoir state-space expansion from temporal bin granularity (Eq. (4)).

Clinical labels and affective categories define the task structure; the representations and their perturbation behavior are the object of study.

II. Related Work

ERP/EEG analysis commonly uses time-window amplitudes and latencies, Hjorth [10] and wavelet/spectral descriptors

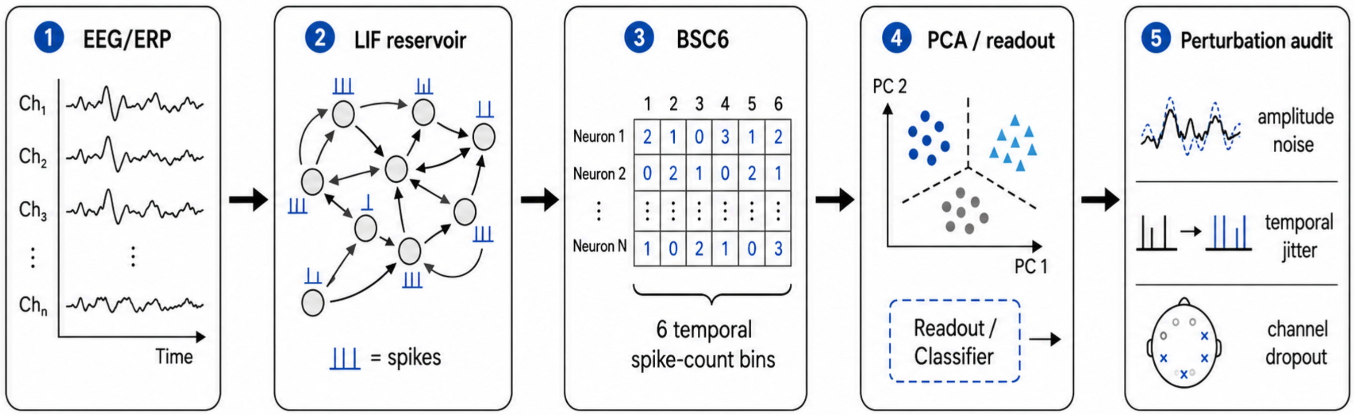

Fig. 1. Reservoir encoding and perturbation-evaluation pipeline. Trial-averaged affective EEG/ERP observations are mapped through a fixed LIF reservoir, summarized by six temporal spike-count bins (BSC_6), compressed for linear readout, and evaluated under amplitude noise, temporal jitter, and channel-dropout perturbations.

[11], and linear models [13]. These features remain attractive because they are interpretable in relation to known ERP components, including the P300 and late positive potential [12], but they can miss nonlinear temporal interactions. Reservoir computing offers a complementary approach: a high-dimensional recurrent dynamical system maps the input to a state trajectory that can be read out by a simple classifier [1]–[4]. Spiking liquid-state machines add event-based timing and count structure [5], [6]. When the recurrent dynamics are fixed, the reservoir can be analyzed as a measurement operator rather than only as a classifier.

Spiking and deep networks are now widely applied to EEG [9], [14], [15]. Much of this literature still emphasizes endpoint accuracy, while preprocessing and acquisition choices receive less systematic treatment [18], [19]. Prior ARSPI-Net work developed neuromorphic reservoir features for affective EEG [7], [8]; the present study narrows the claim to perturbation characterization and scoring artifacts.

Robustness to noise, artifacts, and electrode loss is increasingly studied. Dynamic spatial filtering addresses missing or corrupted channels with a learned channel-attention front end [21], and channel-dropout or amplitude-noise augmentation are standard ways to harden trained models [22]. A recent EEG foundation-model benchmark reports that no single model is most robust to all perturbations and that apparent channel-dropout fragility can depend on whether dropped channels are removed or zero-padded [20]. We build on that methodological point by separating two scoring effects: raw-retention inflation for low-accuracy representations and zero-imputation bias for non-centered fixed features. The fixed reservoir is used as a reference case for this analysis because it is training-free and nearly invariant to the imputation rule.

III. Methods

A. ERP Observation Model

Let $X_s^{(c)} \in \mathbb{R}^{C\times T}$ denote the trial-averaged ERP observation for subject s and affective condition c, with C electrodes (channels) and T temporal samples; observations are stored as $T\times C$ arrays and transposed before encoding. We treat $X_s^{(c)}$ as a noisy observation of an unobserved neural state trajectory:

$$\xi_{t+1} = f_\theta(\xi_t, u_t) + \eta_t, \qquad x_t = H\xi_t + \epsilon_t, \tag{1}$$

where ξ_t is the latent neural state, H is a measurement operator induced by scalp observation, and ϵ_t captures measurement and preprocessing noise. The perturbations below are applied to this observation model (Sec. V-B).

B. Spiking Reservoir Encoding

For each electrode trace, the signal is passed through a fixed LIF reservoir of 256 neurons with recurrent weights W scaled to spectral radius $\rho = 0.9$ (near the edge of stability; Fig. 2d), input weights W_{in}, leak parameter $\beta = 0.05$, and firing threshold ϑ. A compact discrete-time representation is

$$v_{t+1} = \beta v_t + W z_t + W_{\mathrm{in}} x_t - \vartheta r_t, \tag{2}$$
$$z_{t+1} = \mathbf{1}[v_{t+1} \geq \vartheta], \tag{3}$$

where v_t is the membrane state, z_t is the binary spike vector, and r_t denotes reset after threshold crossing. The reservoir is fixed during feature extraction; only the downstream readout is trained.

Post-onset activity is summarized by six equal temporal bins over the window $t \in [10, 70]$ in sample units, which at 256 Hz spans approximately 39–273 ms post-stimulus. For reservoir unit j and bin k, the binned spike-count code is $b_{j,k} = \sum_{t\in\mathcal{B}_k} z_j(t)$, $k = 1,\ldots,6$. Per-electrode spike-bin features are compressed to 64 principal components and concatenated across the 34 channels into a trial-level reservoir representation $E_s^{(c)} \in \mathbb{R}^{2176}$. Writing $s_t = z_t \in \{0,1\}^N$ for the reservoir spike state, the encoding is a fixed composition of a nonlinear temporal map, spike-count binning, and the unsupervised projection:

$$E = \phi_{\mathrm{PCA}}\big([b_1,\ldots,b_6]\big) = \Phi(X), \tag{4}$$

so $E = \Phi(X)$ is a fixed, label-free representation. The PCA basis is fitted once, unsupervised, over spike-bin vectors pooled across all subjects, hence transductive with respect to subject inclusion (bounded by the fold-local control in Sec. V-A). A single-bin control (BSC_1, total post-onset spike count) uses the identical pipeline. The $t \in [10, 70]$ window is early and compact, excluding the later P300/LPP ranges the ERP-window baseline sees, so the clean comparison is conservative for the reservoir.

C. Baselines and Readout

Two classical baselines are evaluated under the same subject-grouped protocol. The band-power baseline (A0) uses spectral band-power features (5 bands $\times$ 34 channels $=$ 170 dimensions). The ERP-window baseline ($\mathrm{A0}_e$) uses mean amplitude in three post-stimulus windows, early (80–200 ms), P300 (250–450 ms), and late positive potential (450–800 ms), per channel ($3 \times 34 = 102$ dimensions). To separate representational robustness from raw dimensionality, we add a dimensionality-matched control: A0 randomly projected to 2176 dimensions (Gaussian random projection, averaged over five random seeds). The band-power, ERP-window, and reservoir representations use a balanced L2-regularized logistic readout to test separability without a high-capacity classifier.

As a trained deep baseline we evaluate EEGNet-8,2 [14] on the raw 34×256 ERP signal under the identical subject-grouped folds, trained per fold (Adam, 80 epochs) with train-only per-channel standardization. EEGNet ingests the raw signal and has no intermediate feature vector, so its perturbations are applied at the input (comparable to the reservoir's raw-signal recomputation, Sec. V-B), whereas the band-power, ERP-window, and reservoir perturbations are feature-level; the cross-family comparison is therefore approximate.

D. Perturbation Diagnostics

Two perturbation regimes are separated. Representation-level perturbations corrupt the extracted feature vector; raw-signal recomputation applies perturbations before feature extraction and recomputes the reservoir representation, propagating corruption through reservoir dynamics. Formally, a perturbation $\mathcal{P}_\eta$ acts either on the representation, $E \mapsto \mathcal{P}_\eta(E)$, or on the observation model of Eq. (1): amplitude noise augments the measurement noise ϵ_t, channel dropout zeroes rows of the measurement operator H, and temporal jitter shifts the sampling index of x_t. We apply additive amplitude noise (signal-to-noise ratios), temporal jitter of the alignment window, and channel dropout at increasing electrode-removal rates. Performance is balanced accuracy (BA) and macro-F1.

E. Robustness Metrics and Two Confounds

We report two robustness summaries. *Raw retention* is the ratio of perturbed to clean BA. Because BA is bounded below by chance ($c = 1/3$ for three balanced classes), raw retention can overstate the stability of representations whose clean performance is already close to chance. We therefore also report *above-chance retention*, $(\mathrm{BA}_{\mathrm{pert}} - c)/(\mathrm{BA}_{\mathrm{clean}} - c)$.

Channel dropout also requires a fill rule for the dropped channel's features. The common zero-fill convention is not neutral after train-only standardization: a zero-valued feature becomes $-\mu/\sigma$, which can be out of distribution for non-centered features. Train-mean imputation maps the dropped feature to the standardized origin. We compare zero-fill with train-mean imputation, then add two in-distribution checks: train-fitted k-nearest-neighbor imputation (k=10) and a within-sample spatial fill using the mean of retained channels. For EEGNet, perturbations are applied to the raw input, and an augmented training condition is evaluated separately.

Non-invariance of zero-imputation robustness. The zero-imputation artifact follows directly from standardization. For a feature with training mean μ and standard deviation σ, an imputed value v enters the readout as $(v - \mu)/\sigma$, so zero- and mean-imputation differ by a fixed shift,

$$\frac{0 - \mu}{\sigma} - \frac{\mu - \mu}{\sigma} = -\frac{\mu}{\sigma}. \tag{5}$$

Train-only standardization leaves clean readout performance unchanged under a constant re-centering of the representation, but the zero-imputation response shifts by b/σ after $x \mapsto x - b$. Thus zero-imputation robustness is not invariant to an information-preserving re-centering. The effect is minimized when $\mu \approx 0$, which is why a mean-centered embedding such as PCA provides a useful reference point for the imputation analysis.

IV. Experimental Protocol

A. Dataset and Task

The experiments use trial-averaged affective ERP data from the Stress, Health, and the Psychophysiology of Emotion (SHAPE) project [24], collected by the Laboratory for Clinical Affective Neuroscience at Stony Brook University, with 211 subjects after exclusion of one anomalous subject record. Each subject contributes three affective conditions (negative, neutral, pleasant), yielding 633 subject-condition observations. The ERP traces contain 34 scalp channels downsampled to 256 Hz, with 256 post-stimulus samples per epoch, baseline-corrected relative to onset; the grand averages show the expected late-positive-potential enhancement for emotional over neutral stimuli (Fig. 3). Subject-level data are maintained in a restricted research environment; only aggregate, de-identified results are reported.

B. Validation

All results are computed under a single protocol: Stratified-GroupKFold (5 folds, seeds 42–46), train-only standardization, and a balanced L2 logistic readout; observations from the same subject never span train and test. We report means with 95% confidence intervals (CIs) over the 25 folds, and macro one-vs-rest AUC from pooled out-of-fold probabilities. Chance is calibrated by a label-permutation null (100 permutations).

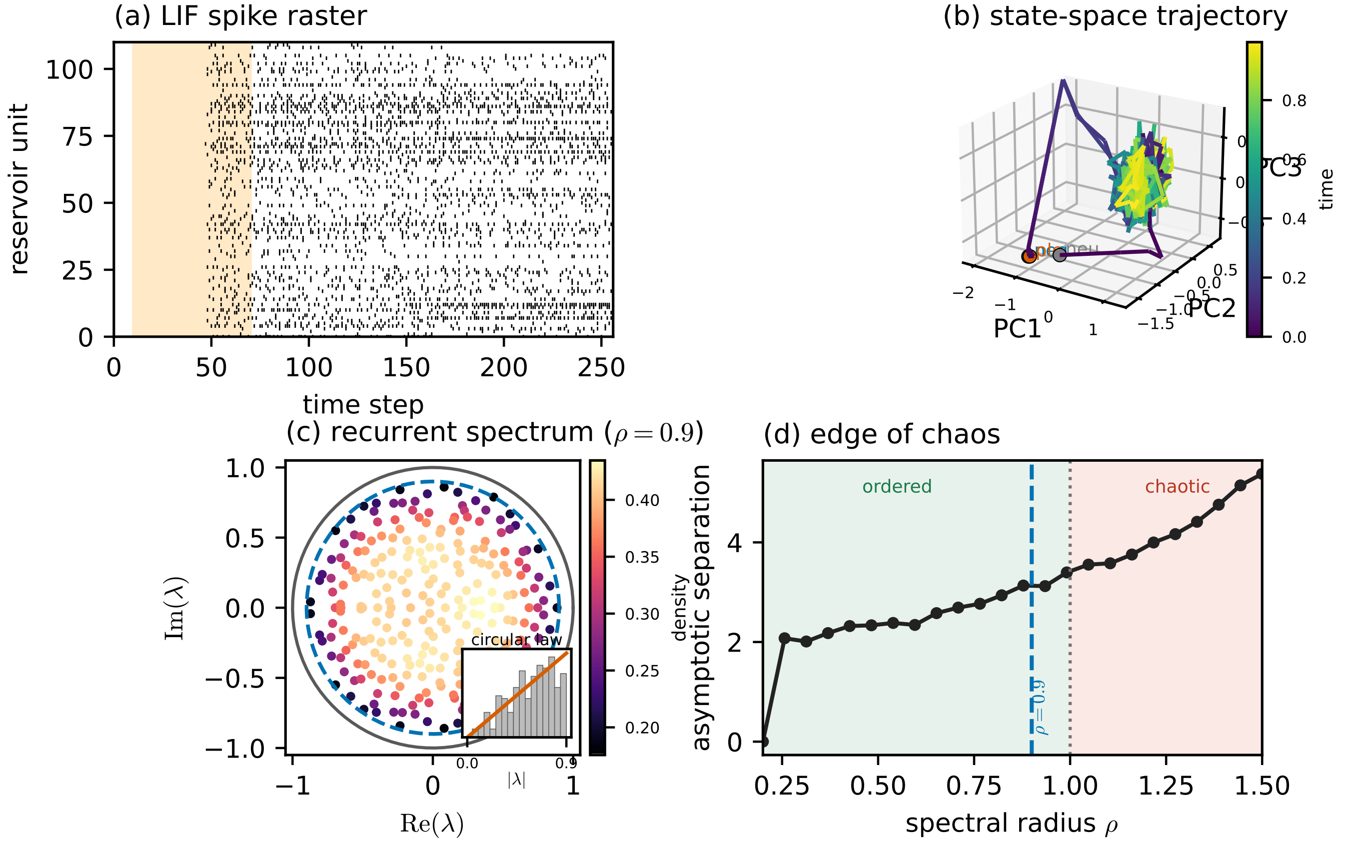

Fig. 2. The fixed LIF reservoir as a dynamical system (montage-free, model-supported). (a) Spike raster (shaded = the $t \in [10, 70]$ analysis window). (b) Population state-space trajectory (first three principal components of the 256-unit membrane state) colored by time, for the three affective conditions; each starts at a marker and settles onto a shared low-dimensional manifold. (c) Density-shaded eigenvalue spectrum of the recurrent weight matrix, filling the disk of spectral radius $\rho = 0.9$ (dashed) within the unit circle in accordance with the random-matrix circular law (inset: radial density vs. the $2r/R^2$ prediction), i.e. operation at the edge of stability. (d) Edge of chaos: asymptotic state separation grows with spectral radius past the critical $\rho = 1$ (dotted); the chosen $\rho = 0.9$ (dashed) sits just below.

V. Results

A. Clean Classification and Controls

Table I reports clean three-class performance. The trained EEGNet is the most accurate representation (BA 0.711); among the fixed encoders, ERP-window features lead ($A0_e$, 0.616), above band-power (A0, 0.485) and the reservoir embedding (A1, 0.463). A label-permutation null gives BA 0.338 ± 0.022, so every configuration encodes genuine signal (all exceed the null by more than five standard deviations) while the absolute clean separability of the fixed encoders remains modest.

Temporal-binning ablation. The single-bin total-count code (BSC_1) performs similarly to BSC_6 on clean classification (0.467 vs. 0.463; Table I) and under perturbation, so the reservoir's behavior is driven by the fixed state-space expansion, not by temporal bin granularity.

Transduction and component controls. A fold-local PCA refit (per-fold training subjects) leaves reservoir performance unchanged (BA $0.463 \to 0.465$, AUC $0.644 \to 0.649$), and varying the component count over $\{16, 32, 64, 128\}$ keeps BA

TABLE I
Clean subject-level performance (mean $\pm$ 95% CI over 25 folds). Permutation-null BA $= 0.338 \pm 0.022$.

Cfg	Features	BA	F1	AUC
D0	EEGNet (raw, trained)	0.711 ± 0.014	0.710	0.868
$A0_e$	ERP-window (102)	0.616 ± 0.016	0.615	0.790
A0	Band-power (170)	0.485 ± 0.015	0.483	0.665
A1	Reservoir BSC_6 (2176)	0.463 ± 0.013	0.460	0.644
–	Reservoir BSC_1 (2176)	0.467 ± 0.011	0.465	0.646

within 0.461–0.491, so the transductive basis and the 64-component choice do not account for the reported results.

B. Two Confounds in Channel-Dropout Robustness

Under *amplitude* noise, the clean accuracy ranking is largely preserved. Degradation is mild for EEGNet and the reservoir (EEGNet $0.711 \to 0.706$, 99% retention; reservoir $0.463 \to 0.449$, 97%), while the classical features fall more (ERP-window $0.616 \to 0.461$; band-power $0.485 \to 0.395$). Channel

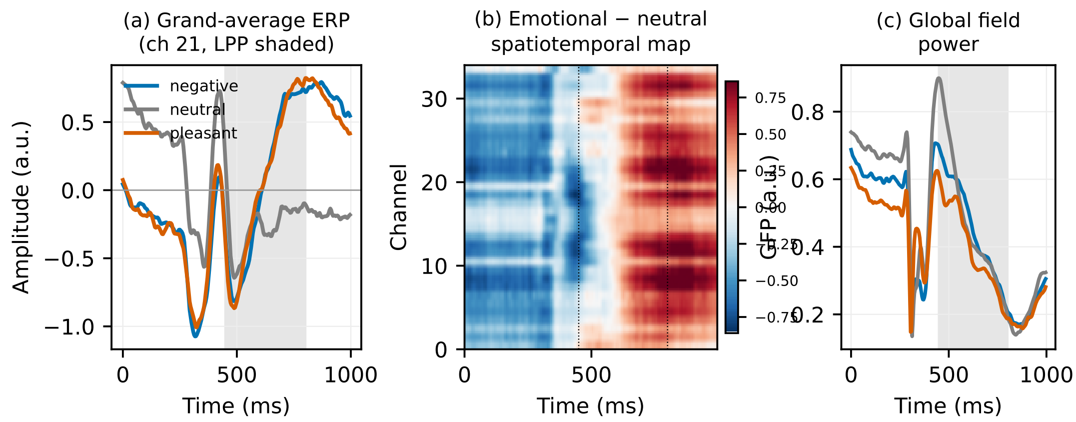

Fig. 3. Trial-averaged affective ERP data (SHAPE cohort; grand averages over 211 subjects). (a) Grand-average waveforms per condition at the most affect-discriminative channel; shaded = the late-positive-potential window (450–800 ms). (b) Spatiotemporal map of the emotional-minus-neutral difference across all 34 channels, showing the sustained LPP enhancement. (c) Global field power per condition.

dropout is less straightforward because the reported ranking changes with the scoring convention (Table II, Fig. 4).

Accuracy-floor effect. With zero-imputation, raw retention ranks the reservoir first at 30% dropout (99% retention). That interpretation is misleading because the reservoir's clean BA (0.463) is close to the $1/3$ chance floor. There is less above-chance performance available to lose. Above-chance retention reduces this advantage and gives a more comparable summary across models (reservoir 95%, band-power 64%, EEGNet 59%, ERP-window 55%).

Zero-imputation effect. Zeroing a dropped channel's features and then standardizing maps them to $-\mu/\sigma$, an out-of-distribution value when the original feature distribution is not centered at zero. Train-mean imputation instead maps the dropped feature to the standardized origin. At 30% dropout, this raises band-power from 0.408 to 0.431 and ERP-window features from 0.454 to 0.489; the largest observed gain reaches +0.043 at 50% dropout. The reservoir changes little (0.456 $\rightarrow$ 0.461), consistent with its PCA-centered embedding. Under zero-imputation, the fixed-encoder order is reservoir, ERP-window, band-power ($0.456 > 0.454 > 0.408$). Under train-mean imputation, the order becomes ERP-window, reservoir, band-power ($0.489 > 0.461 > 0.431$).

Generality. The observed mean-minus-zero gaps at 30% dropout are +0.034 for ERP-window, +0.023 for band-power, and +0.005 for the reservoir, matching the non-centeredness argument in Eq. (5). The pattern holds across 10–50% dropout and across mean, kNN, and spatial fills (Fig. 4). In-distribution fills raise non-centered encoders by up to 0.11 BA, while the reservoir remains within 0.04 BA of its zero-imputation score. At 30% dropout, ERP-window significantly exceeds the reservoir under kNN imputation (0.562 vs. 0.491, p=2×10^{-5}) and spatial fill (p=10^{-3}), is not clearly different under mean fill (p=0.06), and ties under zero-imputation (p=0.8).

Training effect. The unaugmented EEGNet drops to 0.554 BA at 30% channel dropout (78% raw retention). Re-training with channel-dropout and amplitude-noise augmentation raises dropout BA to 0.674 (96% retention) while preserving clean accuracy (0.703). For this trained baseline, the dropout robustness is therefore strongly affected by the training procedure, not only by the architecture (Fig. 5b).

TABLE II

CHANNEL DROPOUT AT 30%: NAIVE AND CORRECTED EVALUATION. "ZERO" AND "MEAN" DENOTE ZERO- AND TRAIN-MEAN IMPUTATION OF DROPPED CHANNELS. RAW-RET. IS $\mathrm{BA}_{\mathrm{zero}}/\mathrm{BA}_{\mathrm{clean}}$; AC-RET. IS ABOVE-CHANCE RETENTION UNDER MEAN-IMPUTATION. EEGNET IS PERTURBED AT ITS RAW, NEAR-ZERO-MEAN INPUT, SO FEATURE IMPUTATION IS NOT APPLICABLE; "+AUG." DENOTES TRAIN-TIME AUGMENTATION.

Representation	Clean	Zero	Mean	raw-ret.	AC-ret.
EEGNet	0.711	0.554	–	78%	59%
+ aug.	0.703	0.674	–	96%	92%
ERP-window	0.616	0.454	0.489	74%	55%
Band-power	0.485	0.408	0.431	84%	64%
Reservoir	0.463	0.456	0.461	99%	98%

Other controls. A dimensionality-matched random projection of band-power (2176-D) drops to 0.399 at 5 dB, similar to band-power and unlike the reservoir. The reservoir's amplitude-noise behavior is therefore not explained by feature dimensionality alone.

Input-level check. Recomputing the reservoir embedding from corrupted *signals*, rather than corrupting the feature vector directly, leaves it relatively stable to amplitude noise and dropout (0.449, 0.431 vs. 0.465 clean) but more sensitive to temporal jitter (0.422).

C. External Replication on an Independent Affective Dataset

We next test whether the imputation effect is specific to SHAPE. On DEAP [23], an independent affective EEG dataset with 32 subjects and 40 trials per subject, we evaluate band-power features on a binary valence task using the standard within-subject protocol (per-subject stratified 5-fold; DEAP band-power is at chance cross-subject). The same pattern

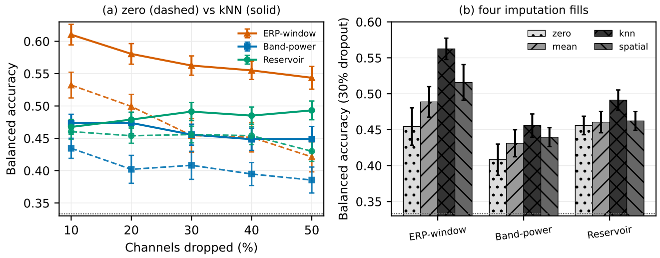

Fig. 4. Channel-dropout performance under different imputation rules. (a) Dropout-rate curves under zero (dashed) and kNN (solid) imputation. The reservoir's centered embedding changes little across fills, whereas ERP-window and band-power features improve under in-distribution fills. (b) At 30% dropout, zero, mean, kNN, and spatial fills move the non-centered encoders by 0.05–0.11 BA, while the reservoir changes only modestly. Dashed line = chance.

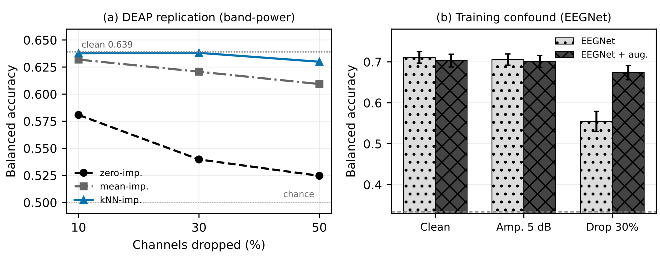

Fig. 5. (a) External check on DEAP [23] (32 subjects, within-subject binary valence, band-power; clean BA 0.639). Channel-dropout BA changes substantially with the imputation rule. (b) EEGNet channel-dropout performance improves from 0.554 to 0.674 when channel-dropout augmentation is used during training, with clean accuracy preserved.

appears, and the magnitude is larger than in SHAPE (Fig. 5a). At 30% dropout, band-power scores BA 0.540 under zero-imputation but 0.621–0.638 under mean or kNN imputation. With clean BA 0.639, zero-imputation indicates a 0.10 BA loss, whereas kNN imputation indicates almost no loss. Thus, for the same features on a second dataset, task, and montage, the imputation convention alone moves measured channel-dropout robustness by up to 0.11 BA.

VI. Discussion

The main result is methodological. Channel-dropout robustness estimates change under two common reporting choices: raw retention can favor models close to the chance floor, and zero-imputation can penalize non-centered features after standardization. For trained models, augmentation is a third factor. After accounting for these choices, the fixed reservoir should not be described as the most channel-robust encoder. Among the fixed encoders, ERP-window features perform best under in-distribution imputation, and the augmented EEGNet is both the most accurate and the strongest dropout baseline in this experiment. The reservoir remains useful as a fixed reference because its centered embedding changes little across imputation rules, making the imputation artifact visible in the other encoders.

These results suggest a minimal reporting standard for EEG robustness studies: report above-chance retention in addition to raw retention, state the channel-fill rule, test at least one in-distribution fill, and report trained-model results with and without perturbation augmentation. None of these additions require a new model, but each changes how the comparison is interpreted.

Limitations. The primary cohort is a single trial-averaged affective-ERP dataset. The imputation effect follows from the standardization algebra in Eq. (5) and is observed across three imputation rules, five dropout rates, and the independent DEAP check, but paradigm-matched multi-cohort ERP validation would still strengthen the claim. EEGNet is perturbed at its raw input while the fixed-feature comparisons are feature-level, so the cross-family comparison is approximate. Adversarial attacks and montage shifts are not evaluated here. Graph topology and structure–function coupling are outside the scope of this paper.

VII. Conclusion

Channel-dropout robustness rankings for affective-ERP representations depend on the scoring protocol. Raw retention can overstate robustness near the chance floor, zero-imputation can distort the behavior of non-centered features after standardization, and trained models need to be evaluated with their augmentation state made explicit. In this study, corrected imputation changes the fixed-encoder ranking, and the same imputation effect appears on DEAP. Future work should test the same reporting protocol on paradigm-matched, multi-cohort ERP data.

Ethics. The SHAPE study was collected by the Stony Brook University Laboratory for Clinical Affective Neuroscience under that laboratory's Institutional Review Board oversight. DEAP is a publicly released dataset governed by its original collection protocol [23].

Data Availability. For information on the SHAPE study dataset, contact the Laboratory for Clinical Affective Neuroscience at Stony Brook University [24]. DEAP is public [23].

References

[1] W. Maass, T. Natschläger, and H. Markram, "Real-time computing without stable states: A new framework for neural computation based on perturbations," *Neural Computation*, vol. 14, no. 11, pp. 2531–2560, 2002.

[2] H. Jaeger, "The echo state approach to analysing and training recurrent neural networks," German National Research Center for Information Technology, GMD Report 148, 2001.

[3] H. Jaeger and H. Haas, "Harnessing nonlinearity: Predicting chaotic systems and saving energy in wireless communication," *Science*, vol. 304, no. 5667, pp. 78–80, 2004.

[4] M. Lukoševičius and H. Jaeger, "Reservoir computing approaches to recurrent neural network training," *Computer Science Review*, vol. 3, no. 3, pp. 127–149, 2009.

[5] P. U. Diehl and M. Cook, "Unsupervised learning of digit recognition using spike-timing-dependent plasticity," *Frontiers in Computational Neuroscience*, vol. 9, p. 99, 2015.

[6] K. Roy, A. Jaiswal, and P. Panda, "Towards spike-based machine intelligence with neuromorphic computing," *Nature*, vol. 575, pp. 607–617, 2019.

[7] A. Lane, W. Tang, and B. Nelson, "Towards ARSPI-Net: Development of an efficient hybrid deep learning framework," in *Proc. IEEE Long Island Systems, Applications and Technology Conf. (LISAT)*, Old Westbury, NY, USA, 2023, pp. 1–8.

[8] A. Lane, W. Tang, and B. Nelson, "Towards ARSPI-Net: Advancing EEG feature extraction with neuromorphic algorithms," in *Proc. IEEE Long Island Systems, Applications and Technology Conf. (LISAT)*, Holtsville, NY, USA, 2024, pp. 1–8.
[9] P. Gong, P. Wang, Y. Zhou, and D. Zhang, "A spiking neural network with adaptive graph convolution and LSTM for EEG-based brain-computer interfaces," *IEEE Trans. Neural Syst. Rehabil. Eng.*, vol. 31, pp. 1440–1450, 2023.
[10] B. Hjorth, "EEG analysis based on time domain properties," *Electroencephalography and Clinical Neurophysiology*, vol. 29, no. 3, pp. 306–310, 1970.
[11] A. Subasi, "EEG signal classification using wavelet feature extraction and a mixture of expert model," *Expert Systems with Applications*, vol. 32, no. 4, pp. 1084–1093, 2007.
[12] S. Makeig, M. Westerfield, T.-P. Jung, S. Enghoff, J. Townsend, E. Courchesne, and T. J. Sejnowski, "Dynamic brain sources of visual evoked responses," *Science*, vol. 295, no. 5555, pp. 690–694, 2002.
[13] B. Blankertz, S. Lemm, M. Treder, S. Haufe, and K.-R. Müller, "Single-trial analysis and classification of ERP components: a tutorial," *NeuroImage*, vol. 56, no. 2, pp. 814–825, 2011.
[14] V. J. Lawhern, A. J. Solon, N. R. Waytowich, S. M. Gordon, C. P. Hung, and B. J. Lance, "EEGNet: A compact convolutional neural network for EEG-based brain-computer interfaces," *Journal of Neural Engineering*, vol. 15, no. 5, 056013, 2018.
[15] A. Craik, Y. He, and J. L. Contreras-Vidal, "Deep learning for electroencephalogram (EEG) classification tasks: A review," *Journal of Neural Engineering*, vol. 16, no. 3, 031001, 2019.
[16] H.-S. Chiang, M.-Y. Chen, and L.-S. Liao, "Cognitive depression detection cyber-medical system based on EEG analysis and deep learning approaches," *IEEE J. Biomed. Health Inform.*, vol. 27, no. 2, pp. 608–616, 2023.
[17] Y. Guo, K. Yang, and Y. Wu, "A multi-modality attention network for driver fatigue detection based on frontal EEG, EDA and PPG signals," *IEEE J. Biomed. Health Inform.*, vol. 29, no. 6, pp. 4009–4022, 2025.
[18] F. Del Pup, A. Zanola, L. F. Tshimanga, A. Bertoldo, and M. Atzori, "The more, the better? Evaluating the role of EEG preprocessing for deep learning applications," *IEEE Trans. Neural Syst. Rehabil. Eng.*, vol. 33, pp. 1061–1070, 2025.
[19] S. Kornblith, M. Norouzi, H. Lee, and G. Hinton, "Similarity of neural network representations revisited," in *Proc. Int. Conf. Machine Learning*, 2019, pp. 3519–3529.
[20] U. Širca, M. Alimardani, S. Zafeiriou, and K. Barmpas, "Beyond accuracy: Robustness, interpretability and expressiveness of EEG foundation models," *arXiv:2605.17562*, 2026.
[21] H. Banville, S. U. N. Wood, C. Aimone, D.-A. Engemann, and A. Gramfort, "Robust learning from corrupted EEG with dynamic spatial filtering," *Journal of Neural Engineering*, vol. 19, no. 4, 046007, 2022.
[22] C. Rommel, J. Paillard, T. Moreau, and A. Gramfort, "Data augmentation for learning predictive models on EEG: A systematic comparison," *Journal of Neural Engineering*, vol. 19, no. 6, 066020, 2022.
[23] S. Koelstra *et al.*, "DEAP: A database for emotion analysis using physiological signals," *IEEE Trans. Affective Computing*, vol. 3, no. 1, pp. 18–31, 2012.
[24] Laboratory for Clinical Affective Neuroscience, "The Stress, Health, and the Psychophysiology of Emotion (SHAPE) project," Stony Brook University, Stony Brook, NY, USA. [Online]. Available: https://lab-can.com